# Significant figures

**1**

**a** Six **b** Four **c** Two **d** Four

**e** Four **f** Five **g** Three **h** Five

**2**

**a** Two **b** Three **c** Three **d** Four

**3** Two

**4** Two

**5** 4549

**6**

**a** 2600 **b** 11 000 **c** 9200 **d** 220

**e** 0.72 **f** 0.030 **g** 5.0 **h** 0.0060

**7**

**a** 1770 **b** 462 000 **c** 6480 **d** 21 000

**e** 9250 **f** 540 **g** 0.563 **h** 0.0736

**8**

**a** 801 600 000 **b** 0.06007 **c** 11.85 **d** 2.160

**e** 4.504 **f** 1.600 **g** 90.03 **h** 0.5987

**9** 7231.0

**10**

**a**

**i** 1000 **ii** 1100

**b** 100

**11**

**a** 3499 **b** 2500 **c** 999

**12** 512

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## Absolute error

**13** **B**

**14** 7.6

**15**

**a** 0.5 kg **b** 0.5 g **c** \$5

**16**

**a** 0.05 mL **b** 0.5 mm **c** 0.05 g **d** 0.5 s

**e** 0.05°F **f** 0.005 kg **g** 0.05 m **h** 0.005°C

**17** $\frac{1}{2}$ mL

**18** We do not know what place value the number has been rounded to.

**19** 11.51 m

**20**

**a**

**i** 3 **ii** 422 $m^3$

**b**

**i** 2 **ii** 2.5 mL

## Limits of accuracy

**21**

**a** 1 cm **b** 90 cm

**c** 0.5 cm **d**

**i** No **ii** No **iii** No **iv** Yes

**22**

**a** 10 kg **b** 5 kg

**c** Lower bound: 65 kg, upper bound: 75 kg

**23**

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**24** 19.995 m

**25** 12.05 m

**26**

**a**
- **i** 13.455 km
- **ii** 13.445 km

**b**
- **i** 6.45 mi
- **ii** 6.35 mi

**c**
- **i** 5.5 yd
- **ii** 4.5 yd

**d**
- **i** 14.005 kg
- **ii** 13.995 kg

**e**
- **i** 4884.5 m
- **ii** 4883.5 m

**f**
- **i** \$52.50
- **ii** \$47.50

**27**

**a** Maximum: 175, minimum: 165

**b** Maximum: 3050, minimum: 2950

**28**

**a** 26.25

**b** 0.6

## Applications

**29**

**a** 1 mm

**b** 79.5 mm

**30**

**a** Upper bound: $272.6925\ \text{m}^2$, lower bound: $269.3925\ \text{m}^2$

**b** Upper bound: 66.2 m, lower bound: 65.8 m

**31** $31.5725\ \text{m}^2$

**32**

**a** 13.85 m/s

**b** 13.95 m/s

**33**

**a** $94.4025\ \text{m}^2$

**b** $92.2625\ \text{m}^2$

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